Aggregation and message passing: standard pool, heuristic, and variational server

The server step lives in `aggregation.py`, where a categorical specialization of the active-inference belief-sharing relation (Friston et al. 2024) and the FedGVI objective (Mildner et al. 2025) meet. Each agent n broadcasts a categorical local posterior $q_n(s)$ over the shared latent factor, optionally with a scalar base weight w_n. Two fusion rules act directly on these broadcasts, and a third — the objective-backed `variational_aggregate` of sec. `sec:method-variational` — refines the second into descent on a stated objective. The first is the **log-linear pool**, a project-local product-of-experts consensus. In the terminology of opinion pooling it is the logarithmic pool, a weighted geometric aggregation rule whose Bayesian-coherence assumptions have been studied independently of active inference (Genest and Zidek 1986; Genest et al. 1986; Carvalho et al. 2023); in machine-learning terms it is the product-of-experts normalization of local posteriors (Hinton 2002):

$$\mathrm{log\_linear\_pool}(\{q_n\}) = \mathrm{softmax}\Big(\sum_n w_n \log q_n\Big), \tag{1}$$

For the source bridge, fix one finite shared support $\mathcal{S}$ with $q_n(s) > 0$ for every agent and state. Suppose the inputs to Eq. 7's softmax message-combination term can be represented as posterior log potentials $m_n(s) = \log q_n(s) + \kappa_n$, where κ_n is constant in s, and use declared fixed weights w_n that do not depend on the emerging consensus (the unweighted case sets each $w_n = 1$). Additive constants then cancel under softmax, giving exactly eq. 1. This is a categorical posterior-log-potential specialization of the source equation's message-combination term, not a reconstruction of source message construction, self-exclusion/cavity policy, scheduling, generative factors, or the complete protocol. The code alias `friston_belief_share` names this qualified specialization only.

The second rule is **`robust_aggregate`**, an iteratively-reweighted pool that discounts each agent by $\exp(-c\,\mathrm{KL}(q_n \,\|\, q))$ against the emerging consensus q. A confidently-wrong (contaminated) agent sits far from the consensus, can earn a small effective weight and be suppressed in the declared diagnostic regimes. This independently motivated rule does not transfer FedGVI's client theorem to the server side: it is the **heuristic robustness axis** of sec. 3.3, distinct from the per-agent rigorous axis of sec. 5.7. It is also only an analogy to robust federated aggregation methods such as divergence-weighted gamma-mean aggregation, geometric-median robust aggregation, or Byzantine-tolerant gradient aggregation (Li et al. 2022; Pillutla et al. 2022; Blanchard et al. 2017): those methods motivate the risk surface, but they do not supply this rule's guarantee.

The defining identity is bit-level: at zero robustness the reweighted pool is the log-linear pool unchanged.

$$\mathrm{robust\_aggregate}(0) \equiv \mathrm{log\_linear\_pool}\,. \tag{2}$$

This is an exact project-local code identity. Under the stated posterior-log-potential assumptions, its right-hand side specializes the message-combination term of Eq. 7; the identity itself neither recovers nor certifies the complete source protocol (Friston et al. 2024).

Protocol map: local updates, broadcast, and server fusion I

The visual map in fig. 1 makes the protocol boundary explicit: each client updates and broadcasts a categorical posterior; the server chooses the standard pool, heuristic, or variational route. This is a mechanistic schematic, not an additional benchmark: client-side FedGVI is source- conditional, server-heuristic accuracy is conditional on declared contamination, and the variational route owns objective/descent/raw-weight properties.

Visual protocol map (schematic) I

Visual protocol map (schematic) I

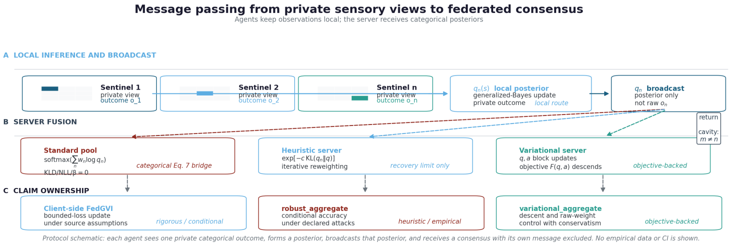

Figure 1: Message-passing schematic for Active Fedference. Source relation: source-inspired original schematic related to Friston et al. (2024), Eq. 7 and Fig. 5; estimand: protocol stages and claim ownership; uncertainty: none. The x-axis is protocol stage from private outcome through local update, posterior broadcast, server fusion, and return; the y-axis uses lanes for local inference, server fusion, and claim ownership. Panel A shows three sentinel agents beginning with private categorical views over the nine-cell location space, converting those views into local posteriors, and broadcasting posteriors rather than raw outcomes. Panel B shows the same broadcast entering the standard log-linear pool, the server-side robust_aggregate heuristic, or the objective-backed variational_aggregate; Panel C keeps their claim ownership separate. The standard pool combines the client KL/NLL/$\beta = 0$ recovery with the qualified categorical Eq. 7 message-combination specialization, while the heuristic retains recovery-limit status only. The return annotation marks cavity exclusion: an agent does not hear its own message. This deterministic formal/mechanistic schematic contains no empirical curve, error band, or confidence interval.

Theorem (Categorical message-combination specialization and local recovery)

Let $\mathcal{S}$ be a finite shared support, let $q_n(s) > 0$ for every n, s, and suppose Eq. 7's softmax inputs are represented by $m_n(s) = \log q_n(s) + \kappa_n$ with κ_n constant in s and fixed declared weights w_n. Then $\mathrm{softmax}(\sum_n w_n m_n)$ equals the log-linear pool of (1). This identifies the categorical message-combination term under those assumptions only. Independently, the project's robust server aggregator at $c = 0$ equals that log-linear pool by (2): every reweighting multiplier is $\exp(0) = 1$, the iteration is skipped, and the same pool code path is returned. Neither statement reproduces the complete source protocol or certifies behavior at $c > 0$.

Corollary (Closed-form Bayes recovery)

With the KL divergence and the NLL loss, the generalized posterior of (1) equals the closed-form prior-times-likelihood Bayes posterior,

$$q^*(s) \;\propto\; \pi_0(s)\,\textstyle\prod_i p(o_i \mid s),$$

so `generalized_posterior(KLD,NLL)` reproduces standard Bayes. Pooling those local posteriors in this project gives the log-linear pool of (1); under the assumptions of Theorem 1, that is the categorical message-combination specialization of Eq. 7, not a recovery of its complete source protocol.

Visual protocol map (schematic) I

The largest observed discrepancy between `robust_aggregate(robustness=0)` and `log_linear_pool` is 0 — bit-identical, since the zero-robustness branch runs the same code path — and between `generalized_posterior(KLD, NLL)` and the analytic Bayes posterior is 5.55e-17, exact to machine precision (about one ULP); both are reported in sec. 7.1, so eq. 2 and eq. 2 are verified identities rather than approximations.

Visual protocol map (schematic) I

The honesty contract binds at exactly this point. The recovery theorem and its corollary cover only the recovery identity and the per-agent rigorous axis of sec. 5.7; no statement *about* `robust_aggregate` transfers a bounded-influence guarantee to that divergence-reweighting, whose positive property is the $\texttt{robustness} = 0$ limit of eq. 2. A scoped no-go rejects a declared separable objective class without supplying a broader objective certificate. The per-agent influence weights that the heuristic produces under contamination are *illustrated*, not guaranteed, in fig. 13 and fig. 16; the genuine per-client FedGVI property is the rcce/AR client loss of sec. 5.7. The next subsection closes this exact gap on the server side with a *different*, objective-backed aggregator.

Variational aggregation with objective-backed weight control I

The related server construction becomes a genuinely variational rule by replacing the heuristic's reverse-KL weight update with a forward cross-entropy update. For $c > 0$ and $\lambda > 0$, treat the consensus q and a vector of effective weights $a = (a_n)$ as joint variational parameters and define the **aggregation free energy**

$$F_\lambda(q, a) = \sum_n a_n \, \mathrm{CE}(q, q_n) - \lambda H(q) + \frac{1}{c} \mathrm{KL}_{\mathrm{gen}}(a \,\|\, w), \tag{3}$$

Variational aggregation with objective-backed weight control I

where $\mathrm{CE}(q, q_n) = -\sum_i q_i \log q_{n,i}$ is the cross-entropy of the consensus relative to agent n, $H(q)$ is the consensus entropy, c is the robustness, and $\lambda > 0$ is the `entropy_weight` coefficient (default $\lambda = 1.0$); $\mathrm{KL}_{\mathrm{gen}}(a \,\|\, w) = \sum_n [a_n \log(a_n/w_n) - a_n + w_n]$ is the generalized KL between the effective and base weights. Each block of F_λ has a closed-form minimizer, so alternating

Variational aggregation with objective-backed weight control I

$$q \leftarrow \mathrm{softmax}\Big(\tfrac{1}{\lambda}\textstyle\sum_n a_n \log q_n\Big), \qquad a_n \leftarrow w_n \exp(-c\,\mathrm{CE}(q, q_n)) \tag{4}$$

is exact block-coordinate descent on eq. 3 (`variational_aggreg ate`) for $c > 0$ and $\lambda > 0$. The implementation defines the $\lambda \downarrow 0$ endpoint separately as a deterministic tied-argmax rule; $\lambda = 0$ is not substituted into the objective or its q-update. This substitution changes both orientation and scale: $\mathrm{CE}(q, q_n) = \mathrm{KL}(q \| q_n) + H(q)$, and its common $H(q)$ term scales all raw weights, which changes the entropy of the subsequent unnormalized weighted log pool. The paired q- and a-updates in eq. 4 are exact block minimizers of the stated objective; that fact does not derive the reverse-KL heuristic. Because F is biconvex, we run the descent

multi-start (pool, uniform, and arithmetic-mean seeds, lowest observed F among converged starts; otherwise the lowest unfinished trace is returned with `converged=False`) so a near-one-hot adversary is not left at the product-of-experts seed in the tested contamination regimes — the detail that supports the effective-weight diagnostic (sec. 11). The full derivation, the formal statement (block descent, $c \to 0$ recovery, and the raw effective-weight bound), and the numerical witnesses are in sec. 11; the empirical descent and influence collapse are shown in fig. 14 and fig. 15 and reported in sec. 7.5.2. The authoritative notation supplement records the complete objective contract in eq. 29.

This upgrades the server side from an untracked heuristic to a derived generalized-Bayes aggregation with an explicit redescending raw-weight property: a single confidently-wrong agent earns raw weight $a_n = w_n \exp(-c\,\mathrm{CE}(q, q_n)) \le w_n$ that vanishes as it diverges, whereas the naive pool grants every agent the fixed weight w_n however wrong it is. The trade is conservatism — the $-H(q)$ term makes the stationary point a maximum-entropy-biased consensus consistent with the weighted cross-entropies, so `variational_aggregate` is deliberately flatter than the product-of-experts and does *not* maximize peak point-accuracy. The two server-side rules therefore play complementary, never-conflated roles, both reported: the sharp `robust_aggregate` heuristic for accuracy under contamination (sec. 7.5.1) and the conservative `variational_aggregate` for a server-side objective with stated weight control (sec. 7.5.2). A temperature parameter $\lambda > 0$ (controlled by `entropy_weight`, default $\lambda = 1.0$) generalizes the objective to F_λ; lower λ sharpens the variational q-block toward a maximizing state for its current weighted log pool. The tempered family (sec. 11.5; objective eq. 25) recovers the full-entropy variational aggregator at $\lambda = 1.0$ and has a separately implemented deterministic tied-argmax endpoint as $\lambda \downarrow 0$; neither endpoint is guaranteed accurate and neither is an algebraic recovery of `robust_aggregate`. The effective-weight a-update is unchanged for every $\lambda > 0$, so the raw-weight bound holds over the objective-defined family.